

Abstract No.

Nuclear adaptation to confined migration fuels glioblastoma aggressiveness

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Context: Glioblastoma (GB) is the most aggressive primary brain tumor, with no curative treatment. During perivascular invasion, GB cells migrate through constrained environments under astrocytic endfeet in response to growth factors. We investigated how mechanical stress influences nuclear envelope (NE) integrity, molecular heterogeneity and GB aggressiveness.

Methods and results: Adherent and patient-derived GB cell lines were not exposed or exposed to an EGF gradient to induce migration in transwell with or without physical constraints (cGB and ncGB). Without EGF, mechanical stress reduced the NE proteins Emerin and SUN1, decreased nuclear circularity, increased micronuclei formation, and H3K9me3 and γ -H2AX foci, suggesting epigenetic and genetic alterations. In contrast, EGF increased lamin A/C expression, preserved nuclear shape, and limited membrane rupture and micronuclei formation under constraint. Using time-lapse videomicroscopy in a PDMS microfluidic device with 3 μ m constrictions, we observed sequential NE bleb formation and lamin A/C "scar" formation, suggesting a dynamic repair network maintaining NE integrity. Proteomic studies of nuclear contents reveal that pathways associated with transcription and spliceosome, metabolic processes and chemokine signaling while RNA-seq confirmed acquired migratory, mesenchymal and immune-related transcriptional programs, specific of confined GB cells. Indeed, these cells exhibited increased proliferation, invasion and resistance to temozolomide. In vivo, immunodeficient mice xenografted with cGB-U87-eGFP+EGF cells showed a 12% decrease in median survival compared with respective ncGB controls. cGB+EGF tumors show increased size and cell density, higher lamin A/C in both tumor core and perivascular infiltrating cells, increased F4/80⁺ macrophages and reduced Iba1⁺/CD68⁺ activated microglia within the bulk GB.

Conclusion: We highlight an original and complex link between physical stress, the expression of NE structural proteins and the protecting role of EGF on the NE integrity during migration under constraint leading to a transition towards an immunosuppressor phenotype and exacerbated aggressiveness likely accompanying recurrence.



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